

swapping it results in paying the individual repo rate of the bond versus receiving SOFR. This is due to the statement above: The bond is a receive coupon versus pay repo position, the asset swap is a pay coupon versus receive SOFR position. Thus, the asset swap spread of US Treasuries can be analyzed by looking at the spread between the repo rate of individual bonds and SOFR. While specialness may cause a difference between the two (unlike in the hypothetical example above), they are usually quite close and hence the asset swap spreads of US Treasuries could be expected to be close to zero – again, speaking from the perspective of different funding rates only.

- Finally, consider asset swapping a German Bund into 3M EURIBOR. This results in a pay repo versus receive 3M EURIBOR basis swap over the life of the bond. When analyzing this asset swap spread, one therefore needs to deal with two types of differences between the funding rates. First, there is a difference regarding the term: While the overnight repo financing matches SOFR, this asset swap involves swapping between overnight and 3M term rates. Second, there is also a difference regarding the type of the funding rate: while both repo and SOFR are secured reference rates, this asset swap involves swapping between a secured and an unsecured rate. As a consequence, the asset swap spread of the German Bund depends on the repo–LIBOR spread and its driving forces. For example, due to the different BIS treatment of secured and unsecured loans, the higher the cost of equity, the lower¹ the asset swap spread should be.

In summary, asset swapping a bond results in a spread position between its repo rate and the reference rate of the swap. One important element of asset swap spread analysis therefore consists in understanding the different types and terms of reference rates. The ‘swap block’ of chapters starts correspondingly with an overview of the reference rates and the driving forces of their spreads in Chapter 11.

This is the foundation to assess the driving forces and fair value of asset swap spreads *from a funding perspective* in Chapter 12. Using the example of the Bund from above, we can determine a fair value for the GC–LIBOR basis swap by considering the different capital requirements among others and then add the specialness component via a repo–GC basis swap. While these basis swaps do not trade, we are in a position to price them and thereby obtain the fair value for the Bund asset swap spread from a funding perspective.

Apart from the funding rate, an asset swap also exchanges the credit exposure of both legs: buying a Bund and asset swapping it into EURIBOR involve both the default risk of Germany and the default risk of LIBOR panel

¹We use the convention Reference Rate + X for the asset swap spread X. For example, when a Bund quotes at EURIBOR –10 bp, the asset swap is –10 bp (not +10 bp). Under this convention, a richening of bonds versus swaps results in lower asset swap spreads.

banks. Under some simplifying assumptions (such as ignoring counterparty risk, haircuts, and margining), one may conceptually consider an asset swap as a set of two swaps: first, a swap between funding rates, and second, a swap between credit risk. Adding this second element to the analysis, one can explain the usually upward-sloping term structure of swap spreads by the default risk of high-quality issuers increasing over time. Moreover, the perspective on asset swaps as incorporating a credit swap naturally establishes a link to credit default swaps (CDS).

At this point, we have formulated the proposition to analyze and price asset swap spreads as a combination of a funding and a credit swap, i.e. via assessing the difference of the funding rates and the credit exposure involved. Do we capture most of the driving forces of asset swap spreads or do we miss important elements? Figure 10.1 answers this question graphically. Verbally: we provide evidence that the credit component does indeed capture several of the influences typically being considered as driving forces of asset swap spreads. More issuance of government debt and factors depending on the economic cycle such as unemployment are correlated to the credit quality of the state. On the other hand, asset swaps are also influenced by factors which are hard to capture via cash-flow analysis and no-arbitrage principles, such as the need of banks to hold bonds to meet capital requirements or to obtain certain benefits. These factors are by nature difficult to observe and to quantify and we address the impact on swap spreads (analysis) in Chapter 18.

The task of analyzing swap spreads becomes even more complex and interesting when considering that all asset swap markets are linked via basis swaps. This results in mutual dependencies and influences of all global bond and swap markets – and means that an individual bond swap spread or an individual basis swap should never be analyzed in isolation. One can draw Figure 10.1 for every currency, as a start, say, for US Treasuries and German Bunds. The funding rates of the two, SOFR and EURIBOR, are exchanged in a cross-currency basis swap (CCBS), which thereby introduces a link between both markets. This has several key consequences:

- One can swap every bond into any reference rate in any currency (and with any coupon payment frequency). Hence, it is insufficient to assess

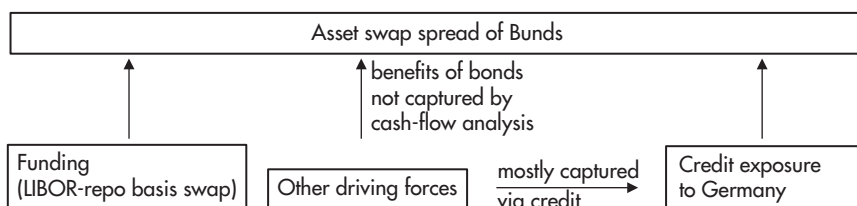


FIGURE 10.1 Driving factors of asset swap spreads.

Source: Authors.

Bunds only versus EURIBOR as they can also be priced versus SOFR. In fact, if basis swapped Bunds appear cheap versus SOFR compared to US Treasuries, it is likely that investors will switch from Treasuries into Bunds, thereby causing them to richen also versus EURIBOR. An analyst who does not consider this link and limits his perspective to EUR has no chance of understanding these moves caused by global asset allocation flows.

- It is possible to use USD SOFR par asset swap spreads of basis swapped bonds as a universal rich/cheap indicator for every bond.

Moreover, the credit element is obviously correlated to the CDS – which adds another link and connects all bond (asset and basis) swap and credit default swap markets worldwide. Figure 10.2 illustrates this statement, as does the following example from our daily experience as analysts: at a time when the SOFR swap spreads of basis swapped JGBs traded close to the Japanese CDS, a tightening of the latter forced JGBs to richen versus SOFR. This enticed investors to switch from basis swapped JGBs into Bunds, causing more negative EURIBOR swap spreads of Bunds. Hence, while neither the EUR funding conditions nor the German credit quality changed, a move in the Japanese CDS had the effect of Bunds richening versus EURIBOR. This cannot be captured by looking at the EUR universe in isolation (Figure 10.1), but only by considering the links to other markets as well (Figure 10.2).

Keeping track of the multitude of links is a complex but worthwhile task, which yields many insights. This is one important goal of the following chapters:

- Since CDS have observable market prices, the idea is to use them in the model for swap spreads as a measure for the credit risk of government bonds. However, due to involving an FX component and in case of physical delivery also a delivery option, sovereign CDS are not a pure measure for credit risk. The foremost task is therefore to extract the pure credit information from the CDS market in order to be able to use it as input in the swap spread formula. Chapter 13 tries to complete this task as well as possible in the current situation and additionally discusses other applications of CDS in RV analysis and trading.
- We then take a closer look at the instruments linking different funding rates: Chapter 14 treats the intra-currency basis swaps (ICBS) between different funding rates in the same currency and Chapter 15 the cross-currency basis swaps (CCBS) between different funding rates in different currencies. Through a set of basis swaps one can swap any funding rate into any other. For example, one can start from SOFR and construct EURIBOR synthetically via a CCBS between SOFR and ESTR and an ICBS between ESTR and EURIBOR.

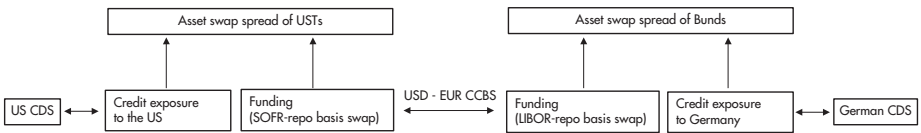


FIGURE 10.2 Mutual influences between the driving factors of asset swap spreads.
Source: Authors.

- We are then prepared to combine the building blocks and to analyze their mutual influences in Chapter 16. By using asset and basis swaps, every bond can be expressed as a spread versus USD SOFR, which is therefore a universal yardstick for comparing all global bonds and the basis for asset allocation and funding decisions. Likewise, the CDS also measures every bond as a spread versus USD SOFR. Using the insights from Chapter 13, we present arbitrage relationships between both. Finally, we combine all relationships into a formula expressing the links between local swap spreads, USD swap spreads, basis swaps, and the CDS.

At the end of this journey, we hope the reader will be in a position to assess the mutual influences between asset swaps, basis swaps, and CDS. This remains a complex problem, as a move in one variable can cause a move in several other variables. In the example above, the switch from basis swapped JGBs into Bunds, which caused a richening of Bunds versus EURIBOR, could have also caused a less negative USD-EUR CCBS (without Bunds richening versus EURIBOR). This situation is somewhat similar to the three-body problem in physics. Fortunately, the arbitrage relationships provide some firm boundaries and orientation.

Chapter 17 presents two options for assessing the relative value between bonds in different currencies. The first method asset and basis swaps every bond in a fixed rate note in a given currency and then applies the fitted curve model from Chapter 8. The second method asset and basis swaps every bond in a SOFR floating rate note, i.e. applies the universal yardstick “SOFR par asset swap spreads of basis swapped bonds” as a universal rich/cheap indicator for bonds. Since only par asset swap spreads with SOFR as the reference rate solve (almost) all deficiencies of asset swap spreads with LIBOR as the reference rate, the second method is only advisable in USD, while the first method can be used in any currency.

All of the above has been based on thinking about asset swap spreads as a combination of a funding and a credit swap (Figure 10.1). While we will provide evidence that this captures many of the actual swap spread moves, some depend on variables, that vary with the circumstances. Haircut funding costs are important examples, which need to be incorporated into swap spread models in order to obtain realistic results (see Figure 12.4) but vary significantly between different market participants. Also important are shadow costs, which are individually different and practically impossible to observe or estimate. Unfortunately, the impact of regulation on markets has generally increased over the last few years, making an already complex task even harder. The final chapter of the ‘swap block’ (Chapter 18) mentions some consequences for swap spread analysis and concludes that, despite all efforts, only partial and approximative predictions are realistic.

Reference Rates

OVERVIEW OF GLOBAL REFERENCE RATES

Until the financial crisis of 2008–2009, the main reference rate in all major markets was LIBOR or a similar rate, i.e. an unsecured term rate. When the credit risk of banks increased during the financial crisis, lenders favored shorter terms in order to minimize credit exposure. The natural endpoint was overnight (O/N) lending and correspondingly a higher volume in overnight indexed swaps (OIS). In addition, following the LIBOR rigging scandal, regulators entered the scene and decided on the reference rate by bureaucratic fiat.¹ Depending on the jurisdiction, the goals and perseverance of regulators were different, leading to different types of reference rates in the major markets:

- In the US, LIBOR was terminated and SOFR (secured overnight financing rate) officially promoted as main reference rate. SOFR is calculated by the Fed based on overnight repo transactions, hence a secured O/N rate. The Fed Funds, an unsecured O/N rate, continues to exist.
- In the eurozone, by contrast, LIBOR was reformed and continues to serve as an unsecured term reference rate. In addition, ESTR, the EUR O/N index average based on unsecured bank lending and being determined by the European Central Bank (ECB), has been introduced as the replacement for EONIA.
- In Japan, the situation is similar to Europe: whereas LIBOR has been terminated, the domestic version, TIBOR, continues to provide an unsecured term reference rate, while TONAR serves as unsecured O/N reference rate.
- In the UK, LIBOR was terminated and SONIA, the GBP O/N index average based on unsecured bank lending and being determined by the BoE, is the only common reference rate.

¹Huggins and Schaller (2022), Introduction.

TABLE 11.1 Most Common Reference Rates in the Major Markets

	Unsecured term	Unsecured O/N	Secured O/N
USD	—	(Fed Funds)	SOFR
EUR	EURIBOR	ESTR	—
JPY	TIBOR	TONAR	—
GBP	—	SONIA	—

Table 11.1 classifies these reference rates by their length (term² or O/N) and type (unsecured or secured). Table 11.1 represents a snapshot of the situation at the time of writing (September 2023), which may evolve in future. Also, as swaps are over-the-counter (OTC) products, other reference rates than those listed can be agreed upon, for example, the ECB policy rate, but these instances are rare and likely to decrease further due to the increasing standardization of swaps by exchanges.

Summarizing the description above in terms of Table 11.1, the starting point was the left column, the unsecured term rate LIBOR. The transition worked generally from the left to the right, but the final goal and the continuation of LIBOR were different in the individual markets. As a consequence, there currently exist three different sorts of reference rates: unsecured term (LIBOR-like) in EUR and JPY, unsecured O/N in all markets, and secured O/N (SOFR) in the US only.

Table 11.1 is the foundation for the discussion of asset and basis swaps in the following chapters:

- Depending on the reference rate used in an asset swap, there are three different types: the fixed coupon payments of a bond can be swapped in LIBOR, or in an unsecured O/N rate such as ESTR, or (in the US) in SOFR. Due to the differences in reference rates relative to the financing rate of the bond (repo), these three swap spreads are different and require different analysis tools. For example, SOFR is by construction closely linked to the repo rate and hence the influence of different funding rates is usually close to zero for asset swap spreads of US Treasuries versus SOFR (i.e. using SOFR as reference rate in the asset swap). In contrast, asset swap spreads of Bunds versus 6M EURIBOR involve both a difference in term (O/N repo versus 6M LIBOR) and in type (secured repo versus unsecured LIBOR). A model for the repo–LIBOR spread is therefore an essential part

²Strictly speaking, also O/N is a “term” with length of one business day. However, we reserve the term “term” for longer periods, such as 1M or 6M.

for analyzing asset swap spreads of Bunds versus EURIBOR and will be presented later in this chapter.

- Intra-currency basis swaps (ICBS) exchange the reference rates listed in different columns of the same row (currency) as well as between different terms of the same cell in the first column (e.g. between 3M and 6M LIBOR). Cross-currency basis swaps (CCBS) exchange the reference rates listed in different rows (currency). Since practically all CCBS use SOFR as one leg and a secured reference rate exists only in the US, CCBS also involve a switch between columns, i.e. between unsecured and secured lending. An understanding of the spreads between different reference rates is therefore also vital for analyzing basis swaps.

Given the classification by the two criteria, term or O/N and unsecured or secured, there are four possible combinations, with one of them, secured term reference rates, being absent from Table 11.1. When US regulators decided to switch to secured reference rates, given the concentration of liquidity in the repo market in O/N, they naturally chose a secured O/N rate, i.e. SOFR. Together with the termination of LIBOR, this resulted in term reference rates being absent from the US market, which posed major difficulties for many market participants and made them reluctant to transition from LIBOR to SOFR. As a compromise between the goal of policy makers (SOFR) and the needs of the market (a term rate such as LIBOR), SOFR term rates became popular. Specifically, the CME Group calculates 1M, 3M, 6M, and 12M term rates from SOFR futures prices. From the viewpoint of the average market participant, such as a corporate treasurer, the reference rate has therefore evolved along the curve depicted in Figure 11.1.

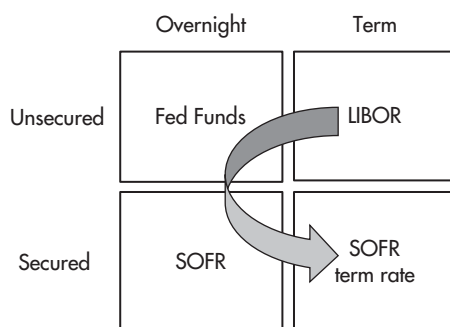


FIGURE 11.1 Evolution of the USD reference rate.

Source: Authors.

While the introduction of a SOFR term rate satisfied the immediate needs of market participants and thereby supported the transition from LIBOR, regulators continue to favor O/N SOFR as reference rate and thus prohibit a secondary market for the term rate. Without a secondary market, the SOFR term rate cannot be practicably used as reference rate for swaps.³ Consequently, our discussion is limited to the three types of asset swap spreads corresponding to the three columns of the reference rate in Table 11.1.

The rest of the chapter first reviews the repo market and SOFR constructed from it, before presenting a model for the spread between repo and unsecured rates.

OVERVIEW OF THE REPO MARKET⁴

A repurchase agreement, or repo, in its most basic form is a type of secured loan. One party lends money for a specified term to a second party, who posts collateral to the first party for the duration of the loan and who pays an agreed interest rate – the repo rate – to the first party on the funds lent. The collateralization by a bond, usually of very high quality like US Treasuries, is different from unsecured lending, such as LIBOR, and tends to result in most observed repo rates being lower than comparable rates involving unsecured, private-sector lending. This difference also leads to repo transactions being assigned lower risk weightings by the Bank for International Settlements (BIS) (0% in the case of US Treasuries) and thereby to lower capital costs. Hence, the motivation for the borrower in the repo market is usually lower interest rates than in unsecured lending markets such as Eurodollar; the motivation for the lender in the repo market is a significant reduction in credit risk and a lower cost of capital.

In addition, the lender can be motivated by receiving a specific bond as collateral in the repo transaction, which he may require to cover a short position in this bond. The fact that the repo markets allows traders to establish short bond positions makes repo a vital component of many relative value trades. As a result, the repo rate appears in many relative value relations and arbitrage equations. For example, the ‘classical’ RV trade of buying a bond

³See Chapter 3 of Huggins and Schaller (2022), for a detailed description and an analysis of the consequences of the prohibition of a secondary market for the SOFR term rate and of scenarios for the future evolution of the tension between policy goals and market needs.

⁴The section is a slightly modified version of the first section of Chapter 1 of Huggins and Schaller (2022).

future versus selling a deliverable bond (net basis) described in Chapter 7 requires establishing and maintaining the short position in the bond until the delivery date of the future – and therefore its repo rate is part of the net basis. Vice versa, relationships between deliverable bonds can be expressed by the implied repo rate, i.e. that repo rate, which would result in a particular bond becoming CTD (cheapest-to-deliver).

The vital function of the repo market for RV trading also means that RV relationships can exert at times a significant influence on the repo market. Using the example of net basis trading, in case of the future trading cheap relative to its CTD, buying the future versus selling the CTD is an attractive arbitrage, whose execution requires establishing the short position in the CTD via the repo market. This situation often results in traders being willing to lend at lower rates, if they receive the CTD as collateral. A borrower in the repo market can then enjoy an especially low rate, if he secures the loan with the CTD rather than general US Treasury collateral. Accordingly, the CTD and its repo rate are called “special,” i.e. below the GC (general collateral) rate offered for loans, in which the borrower can choose any US Treasury as collateral. Apart from CTDs, specialness tends to occur in every bond for which there are significant short positions, which can be caused by RV trading or specific flow situations, such as in the aftermath of a new issuance. Accordingly, specialness tends to appear mainly for new issues, benchmarks, and deliverables, though it can occur in any bond.

If the lender in a repo transaction is unable to return the collateral to the borrower at maturity, this is called a “failure”, which carries penalties. In order to avoid the risk of failures, in addition to the bilateral repo lending described above, tri-party repo can be used, where a custodial bank (as the third party to the repo transaction) holds the collateral. In tri-party repo, no specific bonds can be selected as collateral, so that the tri-party repo rate can be considered as a GC rate with almost no risk of failure. For this reason, the distribution of tri-party repo rates on any given day tends to be narrower than the distribution of bilateral repo rates. Particularly low rates tend to be excluded owing to the absence of special collateral effects and/or the risk of counterparty failures. On the left side, it excludes the special repo rates and any premium charged for the possibility of the counterparty failing.

REPO RATES IN GREATER DETAIL

Unlike LIBOR and OIS rates, repo rates apply to loans that are secured by collateral. While the collateral enhances the security of the loan, it also introduces some additional issues of which the relative value analyst should be aware.